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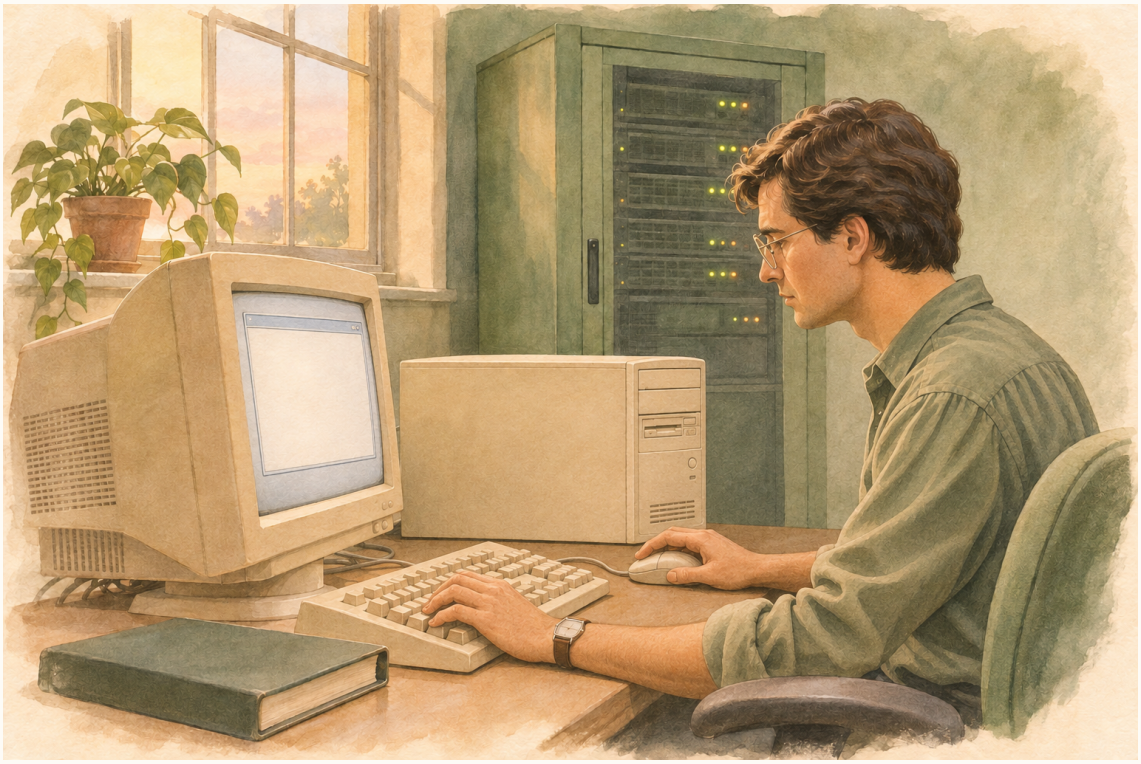
千年虫：被成功抹去的危机

Y2K: The Crisis That Success Erased



危机没有发生，能说明预防多余吗？

英文原文

The Y2K **bug**, or **millennium** bug, began as a **shortcut**. When **memory** and **hardware** were **costly**, **programmers** stored years as two **digits**, writing "99" for 1999. That **compact code** hid an **ambiguity**: a machine might read "00" as 1900, **corrupting** calculations of interest, ages, or **deadlines**.

Governments and firms **braced for chaos**. In February 1998 **President** Clinton created a **council** to **coordinate** America's effort. **Consultants audited legacy** systems and **patched faulty** date logic. In November 1999 the **Commerce Department** estimated American governments and **enterprises** would spend about $100bn; **projections** of the **global** bill ran to hundreds of billions.

Then **midnight struck** on 31 December 1999, and little went wrong. **Glitches** that did **crop up** were **sporadic** and mostly **minor**. Radiation-monitoring equipment **faltered** at Japan's Shika nuclear plant without any threat to safety, and a Pentagon ground system that processes **spy**-satellite data failed. In February 2000 a Senate committee **concluded** that the **expenditure** had been **justified**.

Many saw it differently: the threat had been **exaggerated**, even a **hoax**. Yet calm **prevailed largely** because so much had been fixed. This is the **preparedness paradox**: when **precautions** work, the danger they **ward off** never appears, so it looks **imaginary**. **Tedious** repairs earned **mockery** rather than **gratitude**.

Critics noted that Italy and **South Korea** spent far less yet **fared no worse**. The **comparison misleads**: those low figures often **omitted** what firms spent, and much of the **remedial** work replaced **obsolete** systems **overdue for** upgrading. Nobody can say which dollars were **essential** and which **excessive**. **Prevention** that works may leave nothing but an **invoice**, to **be weighed against** a **catastrophe** that might have been.

中文译文

Y2K 漏洞，又称千年虫，起初只是一条捷径。当年内存和硬件都很昂贵，程序员便把年份存成两位数字，用“99”表示1999年。这种小而紧凑的代码暗藏着一种歧义：机器可能把“00”读成1900年，从而搞错利息、年龄或截止日期的计算。

各国政府和企业都在严阵以待，准备迎接一场混乱。1998年2月，美国总统克林顿设立了一个委员会来协调全美的应对工作。顾问们审查了遗留系统，并给有缺陷的日期处理逻辑打上补丁。1999年11月，美国商务部估算，美国各级政府和企业将为此花费约1,000亿美元；对全球总账单的各种预测则高达数千亿美元。

随后，1999年12月31日的午夜钟声敲响，几乎没出什么问题。确实意外出现的小故障/小毛病也是零星的/偶发的，而且大多轻微。日本志贺核电站的辐射监测设备出了故障，但并未威胁安全；美国国防部一套处理间谍卫星数据的地面系统也曾失灵。2000年2月，美国参议院的一个委员会得出结论，认定这笔开支是合理的。

许多人却不这么看：在他们眼里，这一威胁被夸大了，甚至是一场骗局。然而，平静之所以占据上风，很大程度上正是因为大量问题早已修好。这就是防备悖论：预防措施一旦奏效，它们所抵御的危险就永远不会出现，于是看起来像是虚构的。那些枯燥乏味的修复工作换来的是嘲笑，而不是感激。

批评者指出，意大利和韩国花得少得多，结果也不比别国差。但这种比较具有误导性：那些低支出数字往往遗漏了企业自己的花费，而且很大一部分补救工作替换了早该升级的过时的/淘汰的系统。谁也说不清哪些钱是必不可少的，哪些是过度花费的。奏效的预防，可能只会留下一张账单，用来与一场本可能发生的灾难相权衡。

词汇与表达

bug n.
漏洞
The Y2K bug

millennium n.
千年

shortcut n.
捷径
began as a shortcut

memory n.
内存

hardware n.
硬件

costly adj.
昂贵

programmers n.
程序员

digits n.
数字

compact adj.
小而紧凑的

code n.
代码

ambiguity n.
歧义
That compact code hid an ambiguity

corrupting v.
搞错
corrupting calculations

deadlines n.
截止日期

braced for v.
严阵以待……准备迎接
firms braced for chaos

chaos n.
混乱

President n.
总统

council n.
委员会

coordinate v.
协调
to coordinate America's effort

Consultants n.
顾问

audited v.
审查了
Consultants audited legacy systems

legacy adj.
遗留
legacy systems

词汇与表达（续）

patched v.
给……打上补丁
patched faulty date logic

faulty adj.
有缺陷的

Commerce Department n.
美国商务部

enterprises n.
企业

projections n.
预测
projections of the global bill

global adj.
全球

midnight n.
午夜

struck v.
敲响

Glitches n.
小故障/小毛病

crop up phr.
意外出现

sporadic adj.
零星的/偶发的
sporadic and mostly minor

minor adj.
轻微

faltered v.
出了故障

spy n.
间谍

concluded v.
得出结论

expenditure n.
开支
the expenditure had been justified

justified adj.
合理的
the expenditure had been justified

exaggerated adj.
夸大了
the threat had been exaggerated

hoax n.
骗局
even a hoax

prevailed v.
占据上风

largely adv.
很大程度上

词汇与表达（续）

preparedness paradox n.
防备悖论
the preparedness paradox

precautions n.
预防措施
when precautions work

ward off phr.
抵御
the danger they ward off

imaginary adj.
虚构的

Tedious adj.
枯燥乏味的

mockery n.
嘲笑

gratitude n.
感激

South Korea n.
韩国

fared no worse phr.
不比别国差

comparison n.
比较

misleads v.
具有误导性

omitted v.
遗漏了
omitted what firms spent

remedial adj.
补救

obsolete adj.
过时的/淘汰的
replaced obsolete systems

overdue for adj.
早该

essential adj.
必不可少的
which dollars were essential

excessive adj.
过度花费的

Prevention n.
预防

invoice n.
账单

be weighed against phr.
与……相权衡

catastrophe n.
灾难
a catastrophe that might have been

文章简述

- 隐患：早期电脑为省内存，把1999年只记成“99”。跨入2000年时，“00”可能被当成1900年，导致年龄、利息和截止日期算错，这就是“千年虫”。
- 修补：各国政府和企业提前排查旧系统、修正日期处理。美国估算本国政府和企业总共要花约1000亿美元，全球预计数千亿美元。
- 跨年：2000年到来后，故障大多零星轻微。美国参议院委员会认为投入值得，也有人因此认为先前警告夸大，甚至是骗局。
- 反差：很多问题已提前修好，可能正是没出大事的重要原因；修得越成功，危险越不显眼，反而像原本不存在，这叫“防备悖论”。
- 权衡：不能只因灾难没发生就说钱白花了，要比较修复费用与不修可能造成的损失。但也不能据此断言每笔支出都必要；哪些钱确有用、哪些花多了，仍难逐笔确定。

12

被街道重塑的大脑

The Brain That Learned London

熟记伦敦街道，会怎样改变大脑？

英文原文

Name two addresses and a London **cabbie** will **rattle off** the **quickest route**. That **prowess** is **hard-won**. To earn a **licence**, **aspiring** drivers must **conquer** "the Knowledge", an **arduous examination** on about 25,000 streets within six miles of Charing Cross. **Hopefuls typically** spend three to four years **committing** this **labyrinth** to memory, and only about half **qualify**.

A study published in PNAS in April 2000 **scanned** 16 male cabbies and 50 other men. The **hippocampus** is a seahorse-shaped structure **vital** to memory and **navigation**. In the cabbies, its **rear**, or **posterior**, **portion** was larger than in the other men, and its front, or **anterior**, portion smaller. Longer service went with a larger rear portion.

Yet the link left the cause **uncertain**. Perhaps an **innate flair** for **spatial** tasks **lured** people to the **occupation**, or perhaps **stressful** driving was the **culprit**. A December 2006 study **dispelled** the second **notion**: cabbies had more rear **grey matter**, the tissue **dense** with **nerve** cells, than bus drivers, **confined** to **predetermined** routes, of **equivalent** experience and stress.

A December 2011 **longitudinal** study **tackled** the first, scanning 79 **trainees** and 31 non-trainees at **enrolment** and three to four years later. Their brains began **alike**. Only the 39 who qualified **exhibited** an **expansion** of rear grey matter, so the **sequence** suggests learning **preceded** the **enlargement**.

The gain appeared to **entail** a **trade-off**: qualified drivers **trailed** the non-trainees on one test of recalling an **intricate** figure. The **cohorts** were small and **exclusively** male, yet the studies **imply** the healthy adult brain can **remodel** itself through **intensive** learning. Every street learned, it seems, **occupied** room once **claimed** by something else.

中文译文

随便说出两个地址，伦敦的出租车司机就能脱口说出最快的路线。这份本领来之不易。要拿到执照，有志向的司机必须攻克“知识大考”，这是一场艰巨的考试，考的是约25,000条街道，都在查令十字周边6英里以内。考生们通常要花三到四年把这座迷宫记在脑子里，并且只有大约一半，最终会合格。

一项研究发表于《美国国家科学院院刊》（2000年4月），扫描了16名男性出租车司机和50名其他男性（的大脑）。海马体是一个形似海马的结构，对记忆和导航都是至关重要的。在出租车司机身上，海马体靠后的（也就是后面的）部分比其他男性的更大，而靠前的（也就是前面的）部分则较小。从业时间越长，后部也越大。

然而，这种关联让原因仍然是不确定的。也许是一种天生的、擅长空间任务的天赋把人们吸引到了这个职业里，又或许紧张的驾驶才是罪魁祸首。2006年12月的一项研究打消了第二种想法：出租车司机后部的灰质更多，灰质即密布神经细胞的组织，比公交司机的多；这些公交司机局限于预先确定的线路，经验和压力（与出租车司机）都是相当的。

2011年12月的一项纵向的研究（着手）处理第一种想法（即天赋把人吸引到这一行），扫描了79名学员和31名非学员，分别在入学时和三到四年后（各扫描一次）。他们的大脑一开始都一样。只有最终合格的39人表现出后部灰质的扩大，因此这一先后顺序表明，学习先于（灰质的）增大发生。

这种收获似乎必然带来某种代价：合格的司机落后于非学员，是在“回忆一幅复杂的图形”这项测试上。这些研究的样本群体规模小，而且专门都是男性，但这些研究暗示/意味着，健康的成人大脑可以重塑自身，靠的是高强度的学习。看来，每记住一条街道，都占据了一块空间，这块空间原本被别的东西占用着。

词汇与表达

cabbie n.
出租车司机
addresses and a London cabbie will rattle off the

rattle off phr.
脱口说出

quickest route n.
最快的路线
will rattle off the quickest route.

prowess n.
本领
That prowess is hard-won.

hard-won adj.
来之不易

licence n.
执照

aspiring adj.
有志向的

conquer v.
攻克

arduous adj.
艰巨的
an arduous examination

examination n.
考试

Hopefuls n.
考生们

typically adv.
通常

committing v.
记
committing this labyrinth to memory

labyrinth n.
迷宫

qualify v.
合格
and only about half qualify.

scanned v.
扫描了

hippocampus n.
海马体
The hippocampus

vital adj.
至关重要的

navigation n.
导航
vital to memory and navigation.

rear adj.
靠后的
the cabbies, its rear, or posterior, portion

词汇与表达（续）

posterior adj.
后面的
its rear, or posterior, portion

portion n.
部分

anterior adj.
前面的
its front, or anterior, portion

uncertain adj.
不确定的
Yet the link left the cause uncertain.

innate adj.
天生的
an innate flair for spatial tasks

flair n.
天赋

spatial adj.
空间
spatial tasks

lured v.
吸引

occupation n.
职业

stressful adj.
紧张的

culprit n.
罪魁祸首

dispelled v.
打消了
study dispelled the second notion:

notion n.
想法

grey matter n.
灰质
cabbies had more rear grey matter, the tissue dense with

dense adj.
密布

nerve n.
神经

confined adj.
局限于

predetermined adj.
预先确定的
confined to predetermined routes

equivalent adj.
相当的
of equivalent experience and stress.

longitudinal adj.
纵向的
longitudinal study

词汇与表达（续）

tackled v.
处理

trainees n.
学员

enrolment n.
入学

alike adj.
一样
Their brains began alike.

exhibited v.
表现出

expansion n.
扩大
an expansion of rear grey matter

sequence n.
先后顺序

preceded v.
先于
learning preceded the enlargement.

enlargement n.
增大

entail v.
必然带来
to entail a trade-off:

trade-off n.
代价
appeared to entail a trade-off

trailed v.
落后于

intricate adj.
复杂的

cohorts n.
样本群体
The cohorts were small

exclusively adv.
专门

imply v.
暗示/意味着
yet the studies imply

remodel v.
重塑
can remodel itself

intensive adj.
高强度的

occupied v.
占据了

claimed v.
占用

文章简述

- 训练：伦敦出租车司机通常花三四年记住约2.5万条街道，只有约一半学员通过执照考试。
- 差异：海马体是参与记忆和认路的脑结构；出租车司机的海马体后部比其他男性大、前部小，但最初不知道这是天生差异还是工作造成的。
- 排查：公交司机也要长期驾驶，却只走固定路线；与经验和压力相近的公交司机比较，出租车司机后部富含神经细胞的灰质仍更多。
- 追踪：学员与未参训者起初脑部相似，三四年后只有考过者的后部灰质增加，说明学习发生在大脑变化之前，支持训练可能改变成人大脑。
- 取舍：样本小且全为男性，考过者在一项复杂图形记忆测试中还落后于未参训者；认路能力增强，未必等于各种记忆都增强。

13

两次卖掉埃菲尔铁塔的骗子

The Man Who Sold the Eiffel Tower Twice

铁塔怎么成了骗子的“货物”？

英文原文

In 1925 six **scrap-metal traders** met **secretly** at the Hôtel de Crillon in Paris. A **deputy director-general** of the Ministry of Posts and Telegraphs shared the **momentous** news, **confidentially**, that the Eiffel Tower was to be sold for **scrap**. He invited **tenders** and made each dealer **solemnly vow secrecy**, **ostensibly** to spare the **public indignation**.

The **pretext** was not **absurd**. Built for the 1889 World's Fair, the **monument** had **originally** been **destined** to **come down** after twenty years. The **high-ranking** deputy was Victor Lustig, a **crook** whose **ministerial stationery** was a **forgery**. He had read a newspaper article in which the authorities **lamented** the **hefty cost** of the tower's **upkeep**.

André Poisson, a **businessman impatient** to **gain a foothold** in Paris, won. Lustig **intimated** that his modest **civil-service remuneration** needed **supplementing**, and Poisson paid an **additional bribe**. Lustig left for Vienna with the **takings**.

Fleeced and too **ashamed** to go to the police, Poisson swallowed the loss **quietly**. Not a **whisper** of the **deceit** appeared in print.

About a month later Lustig **had the gall to reappear** with an **encore** of his **script** for new dealers. This time the winner **grew distrustful** and went to the police. The **affair** made the papers, and Lustig made his **getaway**. The **much-repeated** second sale is a **fairy tale**.

Later arrested in the United States for making **fake** money, Lustig broke out of **detention**, was **recaptured** 27 days later and **served time** on the **penal** island of Alcatraz. He died of pneumonia in 1947.

Of Lustig's two marks, the one who paid **kept mum** and the one who kept his money went to the police.

中文译文

1925年，六名经营废金属的商人在巴黎的克里雍酒店秘密地会面了。邮电部的一位副的总负责人向他们私下地分享了一个重大的消息，埃菲尔铁塔将作为废铁出售。他请他们投标，并让每位商人郑重地发誓要保密，表面上是为了避免引起公愤。

这个借口并不荒唐。这座为1889年世博会而建的纪念建筑，原本就注定要在二十年后拆除。那位位高权重的副总长正是维克多·卢斯蒂格，一个骗子，他的那些部长的信笺都是伪造品。他曾读到过报纸上的一篇文章，文中当局哀叹铁塔维护的高昂费用。

安德烈·普瓦松，一个生意人，迫不及待地想在巴黎拥有一席之地，中标了。卢斯蒂格暗示，他那份微薄的公务员薪酬需要补贴，普瓦松然后付了一笔额外的贿赂。卢斯蒂格带着这笔进账去了维也纳。

被骗走钱财，又实在太丢人、没脸去报警，普瓦松只能默默地咽下这笔损失。关于这场骗局的半点风声报纸上没有出现。

大约一个月后，卢斯蒂格厚颜无耻地去再一次出现，对新的一批商人重演他那套剧本。这一次，那位中标者起了疑心，报了警。这桩事件上了报纸，卢斯蒂格逃之夭夭。那个被反复提及的第二次出售，只是个童话。

后来，卢斯蒂格在美国被捕，因为制造假币，他在关押（期间）越狱，27天后又被抓回，并在恶魔岛这座“刑罚之岛”上服刑。1947年，他死于肺炎。

卢斯蒂格的两个目标中，付了钱的那个守口如瓶，保住了钱的那个则去报了警。

词汇与表达

scrap-metal n.
废金属
six scrap-metal traders

traders n.
商人

secretly adv.
秘密地

deputy adj.
副的

director-general n.
总负责人

momentous adj.
重大的

confidentially adv.
私下地
confidentially

scrap n.
废铁
was to be sold for scrap.

tenders n.
投标
He invited tenders

solemnly adv.
郑重地

vow n.
发誓
solemnly vow secrecy

secrecy n.
保密

ostensibly adv.
表面上
ostensibly to spare

public indignation n.
公愤
the public indignation

pretext n.
借口
The pretext was not absurd

absurd adj.
荒唐

monument n.
纪念建筑

originally adv.
原本

destined adj.
注定
destined to come down

词汇与表达（续）

come down v.
拆除

high-ranking adj.
位高权重的

crook n.
骗子
a crook

ministerial adj.
部长的

stationery n.
信笺
ministerial stationery

forgery n.
伪造品
was a forgery

lamented v.
哀叹

hefty cost n.
高昂费用

upkeep n.
维护
the tower's upkeep

businessman n.
生意人

impatient adj.
迫不及待地

gain a foothold v.
拥有一席之地
gain a foothold in Paris

intimated v.
暗示
Lustig intimated

civil-service adj.
公务员

remuneration n.
薪酬
civil-service remuneration

supplementing v.
补贴

additional adj.
额外的

bribe n.
贿赂
an additional bribe

takings n.
进账

词汇与表达（续）

Fleeced v.
骗走
Fleeced and too ashamed

ashamed adj.
丢人……没脸

quietly adv.
默默地

whisper n.
风声

deceit n.
骗局
a whisper of the deceit

had the gall to v.
厚颜无耻地去
had the gall to reappear

reappear v.
再一次出现

encore n.
重演

script n.
剧本

grew distrustful v.
起了疑心
the winner grew distrustful

affair n.
事件

getaway n.
逃之夭夭

much-repeated adj.
被反复提及的

fairy tale n.
童话

fake adj.
假

detention n.
关押
broke out of detention

recaptured v.
抓回

served time v.
服刑

penal adj.
刑罚

kept mum v.
守口如瓶
the one who paid kept mum

文章简述

- 冒充：1925年，卢斯蒂格伪造公文、冒充邮电部官员，邀六名废金属商秘密投标，谎称埃菲尔铁塔要卖废铁。
- 借口：铁塔原拟在世博会20年后拆除，维护费又高；卢斯蒂格利用这些真背景，并以免引公愤为由要求保密。
- 得手：急于在巴黎立足的普瓦松付了购塔款和贿赂，受骗后却羞于报警，让卢斯蒂格带钱逃走。
- 再骗：约一月后，卢斯蒂格重演骗局，新买家起疑报警且未付款；所谓第二次卖成铁塔的故事并不成立。
- 下场：卢斯蒂格后来在美国因造假币被捕，越狱27天后再落网，在恶魔岛监狱服刑，1947年死于肺炎。

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罗马混凝土的千年之谜

The Secret of Rome's Undying Concrete

古罗马混凝土为何如此耐久？

英文原文

Nearly 1,900 years after its **completion**, in about AD 125–128, the Pantheon's **dome** still **crowns** the **vast chamber** below. About 43 metres **in diameter**, it remains the world's largest **unreinforced concrete** dome. It is an **astonishing feat** of **architecture**, and a **stubborn puzzle**.

Modern concrete is usually **reinforced with embedded steel bars**, which are **prone to corrosion**. As the steel **rusts**, it **swells** and **fractures** the concrete around it, and many **structures** are designed for a **lifespan** of around 50 years. Roman concrete is not stronger than its modern **counterpart**; its **merit** lies in **durability**.

Yet the ancient material seemed to have a **shortcoming**. It is **dotted with** white **chunks** of lime, known as lime **clasts**. Experts long **ascribed** these **lumps to sloppy** mixing or **inferior raw ingredients**, a **defect** that, they **assumed**, a more **conscientious** builder would have avoided.

A study published in Science Advances in January 2023 offered a different **interpretation**. The clasts, it suggested, were a **by-product** of "hot mixing" with **quicklime**, a **reactive** lime that **heats up** in water. When water **seeps into** a **crack**, the clasts **dissolve** and their **calcium** recrystallises as calcium carbonate, **plugging** the **gap**. In tests, water **ceased to trickle** through cracked hot-mixed samples **within weeks**, but kept flowing through samples without quicklime.

The **implications extend beyond archaeology**. Cement **manufacture accounts for** about 8 per cent of global carbon dioxide **emissions**, the **think-tank** Chatham House reported in June 2018. Concrete that **mends** its cracks could last longer, making **demolition** and rebuilding rarer and **trimming** emissions. The white lumps once **scorned** as signs of **clumsy** workmanship have **prompted** a **rethink** of construction.

中文译文

万神殿约于公元125至128年竣工，在其竣工近1,900年后，它的穹顶依然位于下方宏大的殿堂的顶部。穹顶直径约43米，至今仍是世界上最大的无钢筋混凝土穹顶。它是建筑史上令人惊叹的壮举，也是一道难以解开的谜题。

现代混凝土通常靠嵌入其中的钢筋加固，而钢筋容易发生锈蚀。随着钢筋生锈，它会膨胀，并使周围的混凝土断裂；而且许多建筑被设计为约50年的使用寿命。罗马混凝土并不比现代的同等对应物更坚固；它的长处在于耐久性。

然而，这种古代材料似乎有个缺点。它的内部散布着白色的石灰小块，称为石灰碎屑。专家长期以来把这些块状物归因于马马虎虎的搅拌或者是比较次的原材料。这种缺陷，他们认为，换一个更加认真负责的建筑工就能避免。

2023年1月发表于《科学进展》（Science Advances）的一项研究提出了不同的解释。这些碎屑，该研究认为，是与生石灰“热混合”的副产品：一种遇水会变热的活性强的石灰。当水渗入裂缝时，碎屑会溶解，它们的钙会重新结晶为碳酸钙，从而堵住缝隙。在试验中，水在几周之内便不再细细流过开裂的热混合样本，但却持续流过不含生石灰的样本。

这项发现的意义超出了考古学的范围。水泥生产占据全球二氧化碳排放量的约8%，智囊团查塔姆研究所于2018年6月报告说。能自行修补裂缝的混凝土可能更经久耐用，从而使拆除与重建减少，并削减排放。这些白色块状物曾被轻视为拙劣的手艺的标志，如今却促使人们重新思考建造之道。

词汇与表达

completion n.
竣工

dome n.
穹顶

crowns v.
位于……的顶部

vast adj.
宏大的

chamber n.
殿堂

in diameter n.
直径

unreinforced concrete n.
无钢筋混凝土

astonishing adj.
令人惊叹的

feat n.
壮举
It is an astonishing feat of

architecture n.
建筑

stubborn adj.
难以解开的

puzzle n.
谜题

reinforced with v.
靠……加固

embedded adj.
嵌入

steel bars n.
钢筋

prone to adj.
容易
which are prone to corrosion.

corrosion n.
锈蚀
which are prone to corrosion.

rusts v.
生锈

swells v.
膨胀

fractures v.
使……断裂

structures n.
建筑

lifespan n.
使用寿命
for a lifespan of around 50 years.

词汇与表达（续）

counterpart n.
同等对应物
stronger than its modern counterpart;

merit n.
长处

durability n.
耐久性
its merit lies in durability.

shortcoming n.
缺点
material seemed to have a shortcoming.

dotted with adj.
散布

chunks n.
小块

clasts n.
碎屑

ascribed to v.
把……归因于

lumps n.
块状物

sloppy adj.
马马虎虎的

inferior adj.
比较次的
mixing or inferior raw ingredients,

raw ingredients n.
原材料

defect n.
缺陷

assumed v.
认为

conscientious adj.
认真负责的
more conscientious builder would have

interpretation n.
解释
offered a different interpretation.

by-product n.
副产品
were a by-product of "hot mixing" with

quicklime n.
生石灰
of "hot mixing" with quicklime,

reactive adj.
活性强的
reactive lime that heats up in water.

heats up v.
变热

seeps into v.
渗入
When water seeps into a crack,

词汇与表达（续）

crack n.
裂缝

dissolve v.
溶解
the clasts dissolve and their calcium

calcium n.
钙

plugging v.
堵住

gap n.
缝隙

ceased to v.
不再

trickle v.
细细流过

within weeks prep.
在几周之内

implications n.
意义
The implications extend beyond

extend beyond v.
超出了……的范围
The implications extend beyond

archaeology n.
考古学

manufacture n.
生产

accounts for v.
占据
Cement manufacture accounts for about

emissions n.
排放
of global carbon dioxide emissions,

think-tank n.
智囊团

mends v.
修补

demolition n.
拆除
making demolition and rebuilding rarer

trimming v.
削减

scorned v.
轻视

clumsy adj.
拙劣的

prompted v.
促使
have prompted a rethink of

rethink n.
重新思考

文章简述

- 耐久：万神殿直径约43米、没用钢筋加固的穹顶已存近1900年。罗马混凝土的优势是能用得久，并非强度高于现代混凝土。
- 疑点：现代钢筋生锈后会膨胀，使周围的混凝土断裂；罗马材料中的白色石灰块，过去却被当成搅拌差或原料劣的缺陷。
- 新解：2023年研究认为，使用生石灰的热混合过程留下这些石灰块；水渗入裂缝后，石灰块溶解，其中的钙重新形成碳酸钙结晶，堵住裂缝。
- 试验：开裂的热混合样本几周内不再渗水，未加生石灰的对照仍漏水，支持材料能自行补缝的解释。
- 前景：水泥生产占据全球二氧化碳排放量的约8%；若能延长建筑寿命，就有望减少拆除、重建及相关排放。

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一夜之间换边开车

The Night Sweden Switched Sides

全国换边行驶，怎样在一个清晨完成？

英文原文

In a 1955 **referendum**, about 83 per cent of voters chose to keep driving on the left, an **overwhelmingly** clear **rejection** of change. On 10 May 1963, however, **legislators** in **parliament overruled** the **electorate**, approving right-hand traffic from 1967 **in defiance of** a **populace** among whom the plan was **unpopular**.

The case rested on **conformity**. **Norway** and **Finland** drove on the right, with about five million **border crossings** a year, so a left-hand **orientation** meant **isolation** and **inconvenient** journeys. Safety favoured **alignment** too. About 90 per cent of cars were left-hand drive, so **motorists overtaking** on narrow two-**lane highways** risked **head-on collisions**.

A state body ran the **mammoth preparation** under a **logo**, a large H with an **arrow**. Its **propaganda** included a song **contest** on television, and the winning **jingle**, whose **lyrics** urged people to keep right, reached fifth in the **chart**. Some 360,000 items of **signage** were changed, and a **massive fleet** of over 1,000 new buses with right-hand doors was **procured**.

The **transition** came on Sunday 3 September 1967. A **curfew** kept **non-essential** vehicles off the **asphalt** from one to six, with essential traffic **exempt**. At ten to five, all traffic stopped and **crawled** across, restarting at five, before **dawn**, with its direction **flipped**. Stockholm and Malmö had a longer **closure**, from Saturday morning to Sunday afternoon, while crews **reconfigured junctions**.

On the following Monday, 125 accidents were reported, against 130 to 198 on earlier Mondays, a **statistical improvement put down to** the **heightened** care of **hesitant** drivers. The **nervousness** apparently **subsided** and **prudent** habits **faded**, for by 1969 accident and **fatality rates** had **reverted to** their **baseline**.

中文译文

在1955年的一次全民公投中，约83%的投票者选择继续靠左行驶，这是一次对改变的压倒性地明确的拒绝。然而1963年5月10日，议会中的立法者推翻了全体选民的意见，批准从1967年起改为靠右行驶。他们全然不顾这里的民众，在民众中这个计划不得人心。

理由在于与邻国保持一致。挪威和芬兰都靠右行驶，每年约有500万车次过境，因此靠左的方向意味着孤立和不便的出行。安全方面也支持统一/一致。约90%的汽车是左舵驾驶，于是驾车者在狭窄的双车道公路上超车时，有发生迎头相撞的危险。

一个国家机关操办了这项巨大的准备工作用了这么一个图标/标识，一个带箭头的大写H。它的宣传包括一场电视歌曲比赛，获胜的广告歌它的歌词劝人们靠右行驶，登上了歌曲排行榜第五名。约36万件标识被更换，并且一支庞大的车队由1,000多辆右侧开门的新公交车组成也已购置到位。

转变于1967年9月3日星期日进行。宵禁令非必要车辆在凌晨1点至6点不得驶上柏油路，必要车辆可以豁免。4点50分，所有车辆停下，缓缓挪到另一侧，5点重新出发，此时还在黎明之前，行驶方向已经翻转。斯德哥尔摩和马尔默的封闭时间更长，从星期六上午持续到星期日下午，工人们在此期间重新设定了交叉路口。

随后的星期一，报告的事故有125起，而此前几个星期一为130至198起；这一统计上的改善被归因于犹豫的驾车人更加的小心。紧张情绪似乎逐渐平息，谨慎的习惯也慢慢消退，到1969年，事故率和死亡率已回到原来的基准水平。

词汇与表达

referendum n.
全民公投
a 1955 referendum

overwhelmingly adv.
压倒性地

rejection n.
拒绝

legislators n.
立法者
legislators in parliament

in defiance of n.
不顾
in defiance of a populace

parliament n.
议会

overruled v.
推翻了
overruled the electorate

electorate n.
全体选民
overruled the electorate

populace n.
民众

unpopular adj.
不得人心

conformity n.
一致
The case rested on conformity

Norway n.
挪威
Norway and Finland drove on

Finland n.
芬兰
Norway and Finland drove on the right,

border crossings n.
过境
about five million border crossings a year, so a left-hand

orientation n.
方向

isolation n.
孤立
meant isolation

inconvenient adj.
不便的

alignment n.
统一/一致
Safety favoured alignment too

motorists n.
驾车者

词汇与表达（续）

overtaking v.
超车
motorists overtaking

lane n.
车道

highways n.
公路

head-on collisions n.
迎头相撞
risked head-on collisions

mammoth adj.
巨大的

preparation n.
准备工作
the mammoth preparation

logo n.
图标/标识

arrow n.
箭头

propaganda n.
宣传

contest n.
比赛

jingle n.
广告歌

lyrics n.
歌词

chart n.
歌曲排行榜

signage n.
标识
360,000 items of signage

massive adj.
庞大的

fleet n.
车队

procured v.
已购置到位
was procured

transition n.
转变
The transition came on Sunday

curfew n.
宵禁

non-essential adj.
非必要

词汇与表达（续）

asphalt n.
柏油路

exempt adj.
豁免
with essential traffic exempt

crawled v.
缓缓挪

dawn n.
黎明

flipped v.
翻转

closure n.
封闭

reconfigured v.
重新设定了
crews reconfigured junctions

junctions n.
交叉路口

statistical adj.
统计上的
a statistical improvement

improvement n.
改善

put down to phr.
被归因于
put down to the heightened care

heightened adj.
更加的

hesitant adj.
犹豫的

nervousness n.
紧张情绪

subsided v.
平息
The nervousness apparently subsided

prudent adj.
谨慎的
prudent habits faded

faded v.
消退

fatality rates n.
死亡率

reverted to v.
回到
had reverted to their baseline

baseline n.
基准
reverted to their baseline

文章简述

- 缘由：瑞典靠左，挪威和芬兰靠右，过境车流频繁；瑞典九成汽车又是左舵车，靠左行驶时司机坐在靠路边的一侧，超车前不易看清对向来车，有迎面相撞风险。
- 决定：1955年公投约83%反对改靠右，议会仍于1963年批准改制，定在1967年实施。
- 准备：政府广泛宣传，更换约36万件道路标识，购置1,000多辆右侧开门的公交车，并在切换期间限制非必要车辆上路。
- 切换：1967年9月3日4点50分，车辆停下后缓缓移到另一侧，5点统一靠右出发；部分城市另封路改造路口。
- 后续：切换后首个周一事故减少，被归因于司机格外谨慎；但到1969年，事故率和死亡率都回到原有水平。

微塑料：无处不在，危害几何?

Microplastics: Everywhere, but How Harmful?

检出了微塑料，就能证明它有害吗?

英文原文

In May 2004 Richard Thompson, a **marine biologist** at the University of Plymouth, and colleagues described an overlooked **pollutant** in the journal Science. **Sifting through plankton** samples kept since the 1960s, they found that **microscopic** plastic **fragments** and **synthetic fibres** had grown more **abundant**. They **labelled** the **particles** microplastics.

Since then the **durable debris** has proved **ubiquitous**, **turning up** on **shorelines**, in deep-sea **sediment**, in **rainfall**, and in Arctic ice. **Regulation** came before proof of harm to people: in December 2015 America **passed a law** banning plastic microbeads from **scrubs** and other **cosmetics rinsed down** drains.

The search **turned inward**. In March 2022 a small Dutch study found plastic in the blood of about three-quarters of 22 healthy **donors**. In March 2024 a study in the New England Journal of Medicine linked plastic in the **fatty plaque** of neck **arteries** to a higher risk of heart attack, **stroke**, or death. Its authors **cautioned** that **correlation** is not **causation**.

The most **sensational** figures are the **shakiest**. In June 2019 a report for WWF, a **conservation** group, **reckoned** people might **swallow** a credit card's worth of plastic a week; critics call that an **overestimate**. Two months later the World Health Organization was **reassuring** but **tentative**: on limited evidence, microplastics in drinking water posed no **apparent** risk at current levels. In February 2025 a study in Nature Medicine reported plastic in brain **tissue** in an amount **popularised as** a **spoonful**; others **queried** its **methodology**.

Detection has outrun understanding; human studies remain small or **observational**. Harm is **plausible** but **unproven**, and until the **verdict** is in, better **measurement** may guide policy more safely than **panic**.

中文译文

2004年5月，理查德·汤普森，一位海洋的生物学家，任职于普利茅斯大学，在《科学》杂志上，他和同事们描述了一种被忽视的污染物。细细筛查自20世纪60年代起保存下来的浮游生物样本，他们发现，极其微小的塑料碎片和合成纤维变得越来越多。他们把这些颗粒称为微塑料。

此后，这些经久不烂的碎屑被证明无所不在，出现在海岸线上、深海沉积物里、雨水中，以及北极的冰层中。监管先于人体受害的证据出现：2015年12月，美国通过了一项法律，禁止把塑料微珠用在磨砂膏等化妆品里，这类产品用后会被冲进下水道。

人们的搜寻随后转向了体内。2022年3月，一项小规模的荷兰研究发现，22名健康献血者中，约四分之三的人血液里有塑料。2024年3月，《新英格兰医学杂志》上的一项研究，将颈部动脉脂肪性斑块中的塑料，与心脏病发作、中风或死亡的更高风险联系了起来。该研究的作者警告称，相关性并不等于因果关系。

最耸人听闻的数字，就是最不可靠的。2019年6月，一份和自然保护组织 WWF 有关的报告估算，人们每周可能吞下相当于一张信用卡分量的塑料；批评者称其高估了。两个月后，世界卫生组织的表态令人安心，但只是试探性的：基于有限的证据，饮用水中的微塑料，按照当前的含量，并未构成明显风险。2025年2月，《自然·医学》上的一项研究报告了脑组织中的塑料，其含量被大众传作“一勺”；其他人则质疑了其研究方法。

检测走在了认识前面；针对人体的研究仍然规模较小，或只是观察性的。说它有害合乎情理，却尚未证实；而在定论出来之前，依据更好的测量来指导政策，或许会更稳妥，胜过任由恐慌左右。

词汇与表达

marine adj.
海洋的

biologist n.
生物学家

pollutant n.
污染物
described an overlooked pollutant in

Sifting through v.
细细筛查
Sifting through plankton samples kept

plankton n.
浮游生物

microscopic adj.
极其微小的
that microscopic plastic fragments and

fragments n.
碎片

synthetic adj.
合成
plastic fragments and synthetic fibres

fibres n.
纤维

abundant adj.
多
fibres had grown more abundant.

labelled v.
把……称为

particles n.
颗粒

durable adj.
经久不烂的
Since then the durable debris has

debris n.
碎屑
Since then the durable debris has

ubiquitous adj.
无所不在
durable debris has proved ubiquitous,

turning up v.
出现

shorelines n.
海岸线

sediment n.
沉积物
on shorelines, in deep-sea sediment,

rainfall n.
雨水

Regulation n.
监管
Regulation came before proof of harm

passed a law v.
通过了一项法律

scrubs n.
磨砂膏

cosmetics n.
化妆品

rinsed down v.
冲进

turned inward adv.
转向了体内

donors n.
献血者
three-quarters of 22 healthy donors.

fatty adj.
脂肪性

词汇与表达（续）

plaque n.
斑块

arteries n.
动脉
in the fatty plaque of neck arteries

stroke n.
中风

cautioned v.
警告

correlation n.
相关性
Its authors cautioned that correlation

causation n.
因果关系
that correlation is not causation.

sensational adj.
耸人听闻的
The most sensational figures are the

shakiest adj.
最不可靠的

conservation n.
自然保护

reckoned v.
估算

swallow v.
吞下

overestimate n.
高估了
critics call that an overestimate.

reassuring adj.
令人安心

tentative adj.
试探性的
was reassuring but tentative:

apparent adj.
明显

tissue n.
组织

popularised as v.
被大众传作
tissue in an amount popularised as a spoonful; others queried

spoonful n.
一勺

queried v.
质疑了

methodology n.
研究方法
others queried its methodology.

Detection n.
检测

observational adj.
观察性的
studies remain small or observational.

plausible adj.
合乎情理
Harm is plausible but unproven,

unproven adj.
尚未证实

verdict n.
定论
and until the verdict is in, better

measurement n.
测量
is in, better measurement may guide

panic n.
恐慌

文章简述

- 来历：2004年，汤普森团队发现海洋样本中微小塑料碎片和合成纤维增多，称其为微塑料。如今从深海到北极都能找到它们。
- 检出：2022年一项小研究在22名健康献血者中，约四分之三的人血中检出塑料。检出颗粒，不能直接证明它已造成疾病。
- 风险：2024年研究发现，颈部血管内脂肪沉积物含塑料的患者，心梗、中风或死亡风险更高；但这只是关联，尚不能证明塑料就是病因。
- 争议：每周吞下“一张信用卡”塑料的估算被批高估；2025年脑组织研究被媒体换算为“一勺”，这是含量估算，不能当作人人脑内的确定含量，测量方法也受质疑。
- 判断：世卫2019年据有限证据，认为饮水微塑料在当时含量下无明显风险。这不等于所有接触都无害；对人体的危害仍需更可靠的研究确认。

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冰里的五千年凶案

The 5,000-Year-Old Murder in the Ice

冰人的箭伤，揭开了怎样的古老谜案？

英文原文

On 19 September 1991 two German **hikers** on a **trek** in the Ötztal Alps found a **corpse** by a **glacier**. It lay on a **ridge** at about 3,210 metres, on the **border** between Austria and Italy. The **assumption** was that a **modern-day climber** had died on an **alpine slope**, since glaciers sometimes yield frozen remains.

Taking it for a recent death, rescuers used a **jackhammer** that **ripped flesh** from his left **hip**. His **possessions** told another story: **leather** and grass clothing, a **copper axe**, a bow and arrows, all **prehistoric artefacts**. Known as Ötzi, the natural **mummy** was a man who lived about 5,300 years ago, between 3350 and 3105 BC.

In 2001, X-rays revealed a **flint** arrowhead **lodged** in his left shoulder, **unseen by examiners** a decade earlier. A **scanner** showed that the **puncture** had **ruptured** an **artery**, a **mortal** injury causing heavy **bleeding**, and left a large **clot**. Researchers believe he died within minutes. He was about 46.

Another person, **it is thought**, killed him. From the wound's depth researchers **inferred** that an **archer** shot him from the **rear**, at about 30 metres. The **shooter** probably **caught** him **unawares**. A 2018 study found he had **dined on** dried ibex meat and fat, **red deer**, and einkorn wheat, a **hearty**, **nourishing** meal reportedly **consumed** within two hours of his death.

Curators in Bolzano now keep Ötzi, whose 61 **tattoos** on **joints** and **spine** were perhaps an early form of **acupuncture**. The archer's **identity** is an **enigma**, and any **motive** is **conjecture**. Ötzi has a name, a last meal, and a cause of death, while his **anonymous killer** has none.

中文译文

1991年9月19日，两名德国徒步者在奥茨塔尔阿尔卑斯山远足时，在一条冰川旁发现了一具尸体。它躺在一道山脊上，海拔约3,210米，位于奥地利与意大利交界处。当时的假设是，一名现代登山者死在了高山的山坡上，因为冰川有时会产出冰冻的遗体。

因为误以为死者刚死不久，救援人员使用风钻（凿冰），（结果）风钻撕下来/扯下了他左侧髋骨的皮肉。他的随身物品却说明了另一回事：皮革和草编的衣物，一把铜制斧子，一张弓和几支箭，全是史前人工制品。被称为奥茨的这具天然木乃伊是一名男子，生活在约5,300年前，也就是公元前3350年至前3105年之间。

2001年，X 光片显示，一枚燧石箭头被卡在了他的左肩里，却未被十年前的检查人员发现。扫描仪显示，这处穿刺伤撕裂了一条动脉，这是会引起大量出血的致命伤，伤处还留下了一大块血块。研究者认为，他在几分钟内就死了。他死时约46岁。

据认为，是另一个人杀死了他。根据伤口的深度，研究者推断，一名弓箭手从背后约30米处向他射了箭。射手很可能趁他毫无防备时出手。2018年的一项研究发现，他吃了干野山羊肉和脂肪、马鹿肉和单粒小麦——一顿丰盛而有营养的饭，据报道是在他死前两小时内吃下的。

博尔扎诺的博物馆馆长如今保管着奥茨，他身上的61处文身位于关节和脊柱，也许是一种早期的针灸形式。那名弓箭手的身份是个谜团，任何动机都只是猜测。奥茨有名字，有最后一餐，也有死因，而杀他的那个不知姓名的凶手这些都没有。

词汇与表达

hikers n.
徒步者

trek n.
远足

corpse n.
尸体

glacier n.
冰川
a corpse by a glacier

ridge n.
山脊
on a ridge at about 3,210 metres

border n.
交界处

assumption n.
假设
The assumption was

modern-day adj.
现代

climber n.
登山者

alpine adj.
高山的

slope n.
山坡

jackhammer n.
风钻

ripped v.
撕下来/扯下了

flesh n.
皮肉

hip n.
髋骨

possessions n.
随身物品
His possessions told another story

leather n.
皮革

copper n.
铜制

axe n.
斧子

prehistoric adj.
史前
all prehistoric artefacts

artefacts n.
人工制品
all prehistoric artefacts

mummy n.
木乃伊
the natural mummy

flint n.
燧石
a flint arrowhead

lodged v.
被卡在了
lodged in his left shoulder

unseen by adj.
未被……发现
his left shoulder, unseen by examiners a decade earlier.

examiners n.
检查人员

scanner n.
扫描仪

puncture n.
穿刺伤
the puncture had ruptured an artery

词汇与表达（续）

ruptured v.
撕裂了
had ruptured an artery

artery n.
动脉
had ruptured an artery, a mortal injury causing

mortal adj.
致命
a mortal injury

bleeding n.
出血

clot n.
血块

it is thought 插入语
据认为

archer n.
弓箭手

inferred v.
推断
researchers inferred

rear n.
背后

shooter n.
射手

caught unawares adv.
趁……毫无防备时出手

dined on v.
吃了

red deer n.
马鹿

hearty adj.
丰盛

nourishing adj.
有营养的
a hearty, nourishing meal

consumed v.
吃下的

Curators n.
馆长
Curators in Bolzano

tattoos n.
文身

joints n.
关节
whose 61 tattoos on joints and spine were perhaps

spine n.
脊柱

acupuncture n.
针灸
an early form of acupuncture

identity n.
身份
The archer's identity

enigma n.
谜团
an enigma

motive n.
动机
any motive is conjecture

conjecture n.
猜测
any motive is conjecture

anonymous adj.
不知姓名的
his anonymous killer

killer n.
凶手

文章简述

- 误判：1991年，两名徒步者在阿尔卑斯山发现冰冻遗体；救援者以为是近期遇难的登山者，使用风钻时还损伤了遗体左侧髋部。
- 身份：皮革和草编衣物、铜斧与弓箭显示他来自史前；这位被称为奥茨的冰人生活于约5300年前，死时约46岁。
- 死因：2001年发现他左肩内有箭头，扫描显示箭伤撕裂动脉；研究者认为他因大量失血，在几分钟内死亡。
- 重建：据伤口推测，射手从背后约30米外放箭；胃中有野山羊肉和脂肪、鹿肉与小麦，据报道是死前两小时内吃下的。
- 未解：他关节和脊柱上的61处文身，可能与早期针灸有关；如今已知道他的最后一餐和死因，射手是谁、为何杀他，却仍无定论。

一颗棉花糖的真相

The Truth About One Marshmallow

一颗棉花糖能预测孩子的未来吗？

英文原文

Picture a **nursery school** on **Stanford**'s **campus** in the late 1960s and early 1970s. A child of about four or five sits **in solitude** before one treat, facing a **blunt bargain**: eat it now, or **resist temptation** for about 15 minutes and **earn** two. **Impulsive** children often **surrender instantly**; **resolute** ones **hold out**.

Follow-ups made the test **legendary**. A 1990 study in Developmental **Psychology** found that, as **adolescents**, longer waiters tended to **attain** higher scores on the SAT, a US **college-entrance exam**. A **trivial snack** became an **emblem** of **virtue**, and **restraint** at four was **glorified** as a **predictor** of **destiny**.

Yet the **foundation** was **precarious**. Only **a few dozen** children entered the SAT analysis, **predominantly** the sons and daughters of Stanford **faculty** and **postgraduate** students. Such a **homogeneous**, **privileged** group could **scarcely** represent **ordinary** families, and **prosperous** homes may **foster** patience and academic success **alike**.

A study published in Psychological Science in May 2018 ran a **conceptual replication**: the same idea, a new sample, **modified** methods. More than 900 **youngsters** from **diverse** backgrounds were **assessed** at four and a half and again at 15. The raw **link** was roughly half as strong as the original finding. **Adjusting for** family background and early **cognitive** ability shrank it by about two thirds, leaving most estimates **statistically insignificant**.

The test is not **worthless**; it may still **reflect** something **substantive**. Research published in Cognition in January 2013 adds **context**: children whose trust an adult had just **betrayed** waited **markedly** less than **peers** dealing with a reliable adult. How long a child **perseveres depends** partly **on** how **trustworthy** the adults around them seem.

中文译文

想象一下一所幼儿园，位于斯坦福大学校园里，时间是20世纪60年代末至70年代初。一个大约四五岁的孩子独自坐在一份零食前，面对一桩直截了当的交易：现在就吃，或者抵挡诱惑约15分钟，就能挣得两份。冲动的孩子往往立刻缴械投降；意志坚定的孩子则坚持抵抗，拒不妥协。

后续研究让这项测验名声大噪（被传为经典）。1990年《发展心理学》上的一项研究发现，到了青少年时期，等得更久的孩子往往在SAT中取得更高分数，（也就是）美国的高考（大学录取考试）。一份微不足道的零食成了美德的象征，而四岁时的克制被美化为命运的预测指标。

然而，这种根据（该研究结论的根基）并不牢靠（站不住脚）。只有几十个孩子进入了SAT分析，主要是斯坦福教师（全体教员）和研究生的子女。如此同质的、优越的群体几乎不能代表普通的家庭，而且富裕的家庭可能既能培养耐心也能促成学业成就（二者兼顾）。

2018年5月发表于《心理科学》的一项研究做了一次概念性的（科研）重复实验：同样的思路，新的样本，调整过的方法。900多名来自多种不同的背景的儿童或青少年在四岁半时接受评估，15岁时再测一次。这个未经加工的关联性，大约只有原来研究结果关联性强度的一半。调整家庭背景和早期认知能力（这些实验变量），缩小了关联性大约三分之二，使得大多数估计值在统计上（完全）微不足道。

这项测验并非毫无价值；它或许仍能反映某些实质性的东西。2013年1月发表于《认知》的研究补充了（一些）背景：刚被成年人辜负信任的孩子，等待的时间明显短于与守信的成年人相处的同龄人。孩子能坚持多久，部分取决于身边的大人看起来有多可信。

词汇与表达

nursery school n.
幼儿园

Stanford n.
斯坦福大学
nursery school on Stanford's campus in the late

campus n.
校园

in solitude n.
独自

blunt adj.
直截了当的

bargain n.
交易

resist v.
抵挡
resist temptation

temptation n.
诱惑
resist temptation for about 15 minutes

earn v.
挣得

Impulsive adj.
冲动的
Impulsive children often surrender

surrender v.
缴械投降
often surrender instantly;

instantly adv.
立刻

resolute adj.
意志坚定的
resolute ones hold out.

hold out phr.
坚持抵抗，拒不妥协
resolute ones hold out.

Follow-ups n.
后续研究

legendary adj.
名声大噪
Follow-ups made the test legendary.

Psychology n.
心理学
in Developmental Psychology found that, as adolescents,

adolescents n.
青少年

attain v.
取得

college-entrance exam n.
高考
on the SAT, a US college-entrance exam.

词汇与表达（续）

trivial adj.
微不足道的

snack n.
零食

emblem n.
象征

virtue n.
美德

restraint n.
克制
restraint at four was glorified

glorified v.
美化
glorified as a predictor of destiny

predictor n.
预测指标
a predictor of destiny.

destiny n.
命运

foundation n.
根据

precarious adj.
并不牢靠
Yet the foundation was precarious.

a few dozen phr.
几十个

predominantly adv.
主要

faculty n.
教师

postgraduate adj.
研究生

homogeneous adj.
同质的
a homogeneous, privileged group

privileged adj.
优越的
a homogeneous, privileged group

scarcely adv.
几乎不

ordinary adj.
普通的

prosperous adj.
富裕的

foster v.
既能培养
foster patience and academic success

alike adv.
二者兼顾

词汇与表达（续）

conceptual adj.
概念性的

replication n.
重复实验
ran a conceptual replication:

modified v.
调整过的

youngsters n.
儿童或青少年

diverse adj.
多种不同的
from diverse backgrounds

assessed v.
评估
were assessed at four and a half

link n.
关联性

Adjusting for v.
调整
Adjusting for family background

cognitive adj.
认知
early cognitive ability

statistically adv.
统计上

insignificant adj.
微不足道
statistically insignificant.

worthless adj.
毫无价值

reflect v.
反映

substantive adj.
实质性的

context n.
背景

betrayed v.
辜负
trust an adult had just betrayed

markedly adv.
明显

peers n.
同龄人

perseveres v.
坚持

depends on v.
取决于

trustworthy adj.
可信
how trustworthy the adults around them

文章简述

- 实验：四五岁孩子面前放一份零食，可以马上吃，也可以独自等约15分钟换两份。研究者记录等待时间，后来再追踪这些孩子。
- 走红：早期追踪发现，等得久的孩子后来在美国大学入学考试SAT中成绩较高，于是这个测试被宣传成未来成功的预测器。
- 局限：早期SAT分析只有数十名孩子，多来自斯坦福大学教职工和研究生家庭。家庭条件可能同时帮助孩子培养耐心、取得好成绩，不能把成绩全归功于自制力。
- 复核：2018年研究追踪900多名不同背景儿童，先发现等待与成绩的关联只有早期的一半；再考虑家庭条件和原有理解、学习能力，关联又减弱约三分之二。结果削弱了原来的推论，却不等于两者毫无关系。
- 信任：2013年实验发现，刚遇到不守信大人的孩子等得更短。等待也受周围环境和成人是否可信影响，不能凭一次测试断定孩子的命运。

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两度摘得诺奖的女人

The Woman Who Won Two Nobels

居里夫人的两座诺奖从何而来?

英文原文

Marie Curie was born Maria Skłodowska on 7 November 1867 in Warsaw, then under Russian **dominion**, where women **were denied** university **admission**. **Eager** to **pursue formal** study, she **emigrated** to Paris in 1891, **enrolling** at the Sorbonne. On 26 July 1895 she **wed** the physicist Pierre Curie, **forging** a research **partnership**.

Henri Becquerel found that uranium emitted **rays**; she named this **property** of certain **elements radioactivity**. In July 1898 the Curies announced polonium, a **tribute to** her **motherland**, and on 26 December, **radium**. In a **leaky** shed they **painstakingly** processed **tonnes** of pitchblende, a uranium **ore**; by 1902 she had **recovered** about a tenth of a **gram** of radium **chloride**.

On 10 December 1903 she **jointly** won the physics Nobel with Pierre and Becquerel, the first woman so **recognised**. The nomination had **initially left** her **out**; at Pierre's **insistence**, the **omission** was **rectified**. On 19 April 1906 a horse-drawn **wagon fatally** struck Pierre. On 5 November she gave her **inaugural** lecture as his **successor**, the Sorbonne's first woman professor.

In November 1911 a press **scandal** about her private life **erupted**; she was urged to stay away. **Undeterred**, she collected the chemistry Nobel, won **outright**, on 10 December. Her **distinction** in two sciences remains **unmatched**. In the First World War she **mobilised** X-ray **vans** and **stationary** posts that examined over a million **casualties**.

She died on 4 July 1934 of a blood **disorder**, probably caused by **prolonged** exposure to radiation. Her notebooks, still **radioactive**, are **preserved** in **lead**-lined boxes at the National Library of France; readers sign a **liability waiver** and **don protective garments**. The pages **retain** her **handwriting** and her radium.

中文译文

玛丽·居里原名玛丽亚·斯克沃多夫斯卡，1867年11月7日生于华沙。当时华沙处于俄国的统治之下，在那里，女性被剥夺了大学的入学资格。渴望追求正规的学习，她于1891年移居巴黎，注册入读索邦大学。1895年7月26日，她与物理学家皮埃尔·居里成婚，两人建立起研究上的合作关系。

亨利·贝克勒尔发现，铀会发出射线；她把某些元素的这种性质/属性命名为放射性。1898年7月，居里夫妇宣布发现了钋，以此致敬她的祖国；12月26日，他们又宣布发现了镭。在一间漏雨的棚屋里，他们费尽心力地处理了成吨的沥青铀矿，这是一种铀矿石。到1902年，她已提取出约十分之一克的氯化镭。

1903年12月10日，她与皮埃尔和贝克勒尔共同获得诺贝尔物理学奖，第一位获此认可的女性。提名最初把她排除在外了；在皮埃尔的坚持下，这一疏漏得到了纠正。1906年4月19日，一辆马拉的货车致命地撞上了皮埃尔。11月5日，她作为他的继任者发表了就职演讲，成为索邦大学第一位女教授。

1911年11月，一场关于她私生活的媒体丑闻爆发，有人劝她不要前往。她并没有被吓住，于12月10日，领取了完全属于她的诺贝尔化学奖。她在两门科学中双双获得诺贝尔奖，这份荣誉至今仍无人企及。第一次世界大战期间，她动员/调用了X光面包车和固定站点，这些设施检查了超过一百万伤亡人员。

1934年7月4日，她死于一种血液疾病，这很可能是由长期暴露于辐射所导致的。她的笔记本至今仍有放射性，保存在法国国家图书馆的铅衬的盒子里；读者须签署免责协议，并穿上防护服装。这些书页留存着她的笔迹，同样也留存着她所发现的镭。

词汇与表达

dominion n.
统治
then under Russian dominion,

were denied v.
被剥夺了
were denied university admission.

admission n.
入学资格
were denied university admission.

Eager adj.
渴望

pursue v.
追求
Eager to pursue formal study,

formal adj.
正规的

emigrated v.
移居
she emigrated to Paris in 1891,

enrolling v.
注册入读
emigrated to Paris in 1891, enrolling

wed v.
成婚

forging v.
建立起
forging a research partnership.

partnership n.
合作关系
forging a research partnership.

rays n.
射线

property n.
性质/属性
she named this property of certain

elements n.
元素

radioactivity n.
放射性
of certain elements radioactivity.

tribute to n.
致敬

motherland n.
祖国

radium n.
镭

leaky adj.
漏雨的

painstakingly adv.
费尽心力地
they painstakingly processed tonnes of

tonnes n.
吨

ore n.
矿石

recovered v.
提取
by 1902 she had recovered about a

gram n.
克

chloride n.
氯化

jointly adv.
共同
10 December 1903 she jointly won the

recognised v.
获此认可的

initially adv.
最初

词汇与表达（续）

left out v.
把……排除在外了

insistence n.
坚持
Pierre's insistence, the omission was

omission n.
疏漏
Pierre's insistence, the omission was

rectified v.
纠正
the omission was rectified.

wagon n.
货车

fatally adv.
致命地

inaugural adj.
就职
she gave her inaugural lecture as his

successor n.
继任者
inaugural lecture as his successor,

scandal n.
丑闻

erupted v.
爆发

Undeterred adj.
没有被吓住
Undeterred,

outright adv.
完全

distinction n.
荣誉
Her distinction in two sciences

unmatched adj.
无人企及

mobilised v.
动员/调用了
World War she mobilised X-ray vans and

vans n.
面包车

stationary adj.
固定

casualties n.
伤亡人员

disorder n.
疾病

prolonged adj.
长期
probably caused by prolonged exposure

radioactive adj.
有放射性

preserved v.
保存

lead n.
铅

liability waiver n.
免责协议

don v.
穿上

protective adj.
防护

garments n.
服装

retain v.
留存

handwriting n.
笔迹

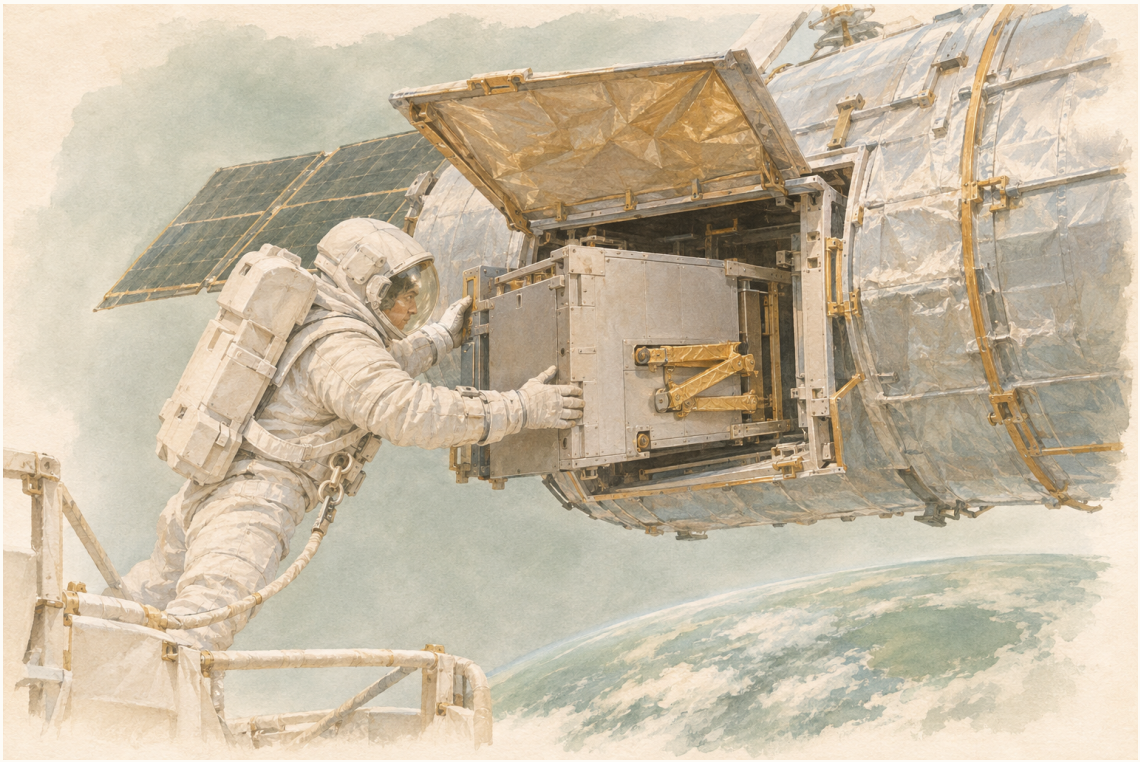
文章简述

- 求学：当时华沙的大学不收女性，玛丽·居里1891年赴巴黎求学，后与皮埃尔结婚合作研究。
- 发现：贝克勒尔发现铀会放出射线，她将这种性质称为放射性；居里夫妇发现钋、镭，从成吨矿石提得约0.1克氯化镭。
- 首奖：1903年，她与皮埃尔、贝克勒尔共享物理学奖，成为首位女性诺奖得主；最初漏提名经皮埃尔坚持纠正。
- 双奖：皮埃尔遇难后，她接任成为索邦首位女教授；1911年顶住私生活风波独获化学奖，仍是唯一获两门科学诺奖者。
- 救护代价：一战时她组织X光设施检查逾百万伤员；后来死于很可能与长期辐射有关的血液病，笔记至今仍有放射性。

20

需要戴眼镜的望远镜

The Telescope That Needed Glasses



主镜没有修好，影像怎样变清晰？

英文原文

On 24 April 1990 the space shuttle Discovery **lofted** the Hubble Space **Telescope into** orbit. The **observatory cradled** in the **payload** bay had cost about $1.5 billion, a **lavish investment** meant to fix its **gaze** on the **cosmos** and its **galaxies**. Money, it soon emerged, could not buy **clarity**.

Its first **snapshots** came back **blurred** and **fuzzy**. In June 1990 NASA announced a **flaw** in the main mirror: its **outer rim** had been **flattened** by 2.2 micrometres, roughly a fiftieth of the **width** of a hair. That **deviation marred** the **optics**, leaving light without a **sharp** focus.

The trouble began with **calibration**. A **lens** in the **assembly** that checked the mirror's shape had been **offset** by about 1.3 **millimetres**, so the mirror **was ground to** the wrong **curve**. A second, **independent test** by the **contractor**, Perkin-Elmer, gave a **divergent** result, but engineers and NASA managers judged it less **reliable** and left the **contradictory** reading **unexplored**.

The mirror could not be repaired in orbit, so engineers **prescribed tailored** glasses. COSTAR, a **corrective cabinet** the size of a **refrigerator**, carried small mirrors on **adjustable** arms that **intercepted** the light and **compensated for** the **imperfect** mirror's **distortion** before the light reached the **instruments**. The mirror itself would stay **untouched**, and a new main camera was prepared.

On 2 December 1993 Endeavour **ascended** with seven astronauts. After a **rendezvous**, they worked **in tandem** across five **spacewalks**, about 35 and a half hours of **exertion**, **installing** COSTAR and the new camera. The **imagery** turned **crisp**, Hubble's **vision renewed**. The mirror still carries that 2.2-micrometre flaw, and the light it **gathers** is corrected only after leaving it.

中文译文

1990年4月24日，发现号航天飞机把哈勃空间望远镜送上轨道。这座安放在有效载荷舱里的天文台耗资约15亿美元，是一笔巨大的/奢侈的投资，旨在把目光投向宇宙及其中的星系。人们很快发现，金钱买不来清晰度。

它最初的快照传回来时模糊不清、很不清楚。1990年6月，美国航天局宣布主镜的一个缺陷，主镜外侧的边缘被磨平了2.2微米，大约相当于头发丝宽度的五十分之一。这一偏差损害了光学系统，使光线无法形成锐利的焦点。

问题始于校准。检测主镜形状的那套组件中，一块透镜错位了/偏离了约1.3毫米，主镜因此被磨成了错误的曲面。承包商珀金-埃尔默公司做的第二次独立测试给出了不同的结果，但工程师和美国航天局的管理人员认为它不太可靠，这一相互矛盾的读数也就无人深究。

主镜无法在轨道上修复，于是工程师开出了一副量身定制的眼镜。COSTAR 是一个冰箱大小的矫正箱体，可调节的机械臂上装着小镜子，截住光线，抵消有缺陷的主镜造成的畸变，这些都需要在光线到达仪器之前完成。主镜本身将保持原封不动，一台新的主相机也已准备就绪。

1993年12月2日，奋进号载着七名宇航员升空。完成交会后，他们协同进行了五次太空行走，辛苦作业共约35个半小时，安装了COSTAR 和新相机。影像变得清楚分明，哈勃的视力得以恢复。主镜至今仍带着那2.2微米的缺陷，它收集的光要离开镜面之后才会被矫正。

词汇与表达

lofted into v.
把……送上

Telescope n.
望远镜

observatory n.
天文台
The observatory

cradled v.
安放

payload n.
有效载荷
in the payload bay

lavish adj.
巨大的/奢侈的
a lavish investment

investment n.
投资

gaze n.
目光

cosmos n.
宇宙

galaxies n.
星系

clarity n.
清晰度
could not buy clarity

snapshots n.
快照

blurred adj.
模糊不清

fuzzy adj.
很不清楚

flaw n.
缺陷
a flaw in the main mirror

outer adj.
外侧的

rim n.
边缘

flattened v.
磨平了

width n.
宽度

deviation n.
偏差
That deviation marred the optics

词汇与表达（续）

marred v.
损害了

optics n.
光学系统
marred the optics

sharp adj.
锐利的

calibration n.
校准
The trouble began with calibration

lens n.
透镜

assembly n.
组件
A lens in the assembly

offset v.
错位了/偏离了
offset by about 1.3 millimetres

millimetres n.
毫米

was ground to v.
被磨成了……

curve n.
曲面

independent test n.
独立测试

contractor n.
承包商
the contractor, Perkin-Elmer

divergent adj.
不同的
a divergent result

reliable adj.
可靠
judged it less reliable

contradictory adj.
相互矛盾的
the contradictory reading

unexplored adj.
无人深究

prescribed v.
开出了

tailored adj.
量身定制的

corrective adj.
矫正
a corrective cabinet

词汇与表达（续）

cabinet n.
箱体

refrigerator n.
冰箱

adjustable adj.
可调节的

intercepted v.
截住
intercepted the light

compensated for v.
抵消
compensated for

imperfect adj.
有缺陷的

distortion n.
畸变
the imperfect mirror's distortion

instruments n.
仪器
light reached the instruments.

untouched adj.
原封不动
The mirror itself would stay untouched

ascended v.
升空

rendezvous n.
交会
After a rendezvous

in tandem n.
协同
worked in tandem

spacewalks n.
太空行走

exertion n.
辛苦作业

installing v.
安装了

imagery n.
影像
The imagery turned crisp

crisp adj.
清楚分明

vision n.
视力

renewed v.
恢复

gathers v.
收集

文章简述

- 失焦：1990年，耗资约15亿美元的哈勃望远镜升空，传回的照片却模糊。主镜边缘磨得过平，误差只有2.2微米，约为头发宽度的五十分之一，却让光线无法形成清晰的焦点。
- 错检：地面检测装置里一块透镜错位约1.3毫米，主镜因此按错误的曲线打磨。另一项独立测试早给出不同结果，但工程师和美国航天局管理者认为它不够可靠，没有继续追查。
- 办法：主镜无法在轨道上修复，工程师便给哈勃配“眼镜”：冰箱大小的COSTAR箱体伸出机械臂，用小镜子截住主镜反射的光，抵消畸变，让光线进入观测仪器。
- 维修：1993年12月，七名宇航员乘奋进号升空，通过五次太空行走，协同工作约35个半小时，装上COSTAR和一台新的主相机。
- 结果：照片恢复清晰，哈勃重新具备观察宇宙的能力；但原主镜没有被重磨，2.2微米的缺陷仍在。修复的关键，是让光线离开主镜后、进入仪器前得到矫正。